Lecture 28: Taylor and Maclaurin Series (II) (10.7)

$c_n = \frac{f^{(n)}(a)}{n!}$

ex. 1. (a) Find the Maclaurin series and $= \sum_0^\infty c_n x^n$

(b) Taylor Series centered at $a = 2$ for $= \sum_0^\infty c_n (x-2)^n$

$$p(x) = x^3 - 2x^2 + x - 3.$$

(c) Use both series to find $p(2.1)$.

(d) Which series is easier to use to evaluate $p(2.1)$?

M. series. T. series.

n	$P^{(n)}(x)$	$P^{(n)}(0)$	c_n	$P^{(n)}(2)$	c_n
0	x^3-2x^2+x-3	-3	-3	$(2)^3-2(2)^2+2-3=-1$	-1
1	$3x^2-4x+1$	1	1	$3(2)^2-4(2)+1=5$	5
2	$6x-4$	-4	$-4/2! = -2$	$6(2)-4=8$	4
3	6	6	$6/3! = 1$	6	1
4	0	0		0	
$\vdots$	$\vdots$	$\vdots$		$\vdots$	
n	0	0		0	

$c_n = 0$, $n \geq 4$

M. series (a) $P_{a=0}(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots = -3 + 1\cdot x + (-2)x^2 + 1\cdot x^3$

$$P_{a=0}(2.1) = \underbrace{-3 + (2.1)}_{\sim -0.9} - 2(2.1)^2 + (2.1)^3 = -0.459$$

T. series (b) $P_{a=2}(x) = -1 + 5(x-2) + 4(x-2)^2 + (x-2)^3$

$$(= -1 + 5x - 10 + 4(x^2 - 4x + 4) + x^3 - 6x^2 + 12x - 8)$$

$$(= -3 + x - 2x^2 + x^3 = P(x))$$

$$P_{a=2}(2.1) = \underbrace{-1 + 5(2.1-2)}_{\sim -0.5} + 4(2.1-2)^2 + (2.1-2)^3 = \boxed{-0.459}$$ ← easier

better approx.

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ex. 2. (i) Find the Taylor Series for $f(x) = \dfrac{1}{x} = \sum c_n (x-1)^n$ centered at $a = 1$. Write your answer in summation notation.
What is the radius of convergence?

n	$f^{(n)}(x)$	$f^{(n)}(1)$	$c_n = \dfrac{f^{(n)}(1)}{n!}$
0	$\frac{1}{x} = x^{-1}$		
1	$(-1)x^{-2}$		
2	$(-1)(-2)x^{-3}$		
3	$(-1)(-2)(-3)x^{-4} = \frac{(-1)^3 3!}{x^4}$		
⋮			
n	$\frac{(-1)^n n!}{x^{n+1}}$ $n \geq 0$	$\frac{(-1)^n n!}{1^{n+1}}$	$\frac{(-1)^n \cancel{n!}}{\cancel{n!}} = (-1)^n = c_n$

$$\frac{1}{x} = \sum_{n=0}^{\infty} (-1)^n (x-1)^n$$

(Root test $\longrightarrow$ $R = 1$) $\rightarrow$ check end pts.

$$= \sum_{n=0}^{\infty} \underbrace{[-(x-1)]^n}_{ar^n}$$

geometric series $\left| r = -(x-1) \right| < 1$

$|x-1| < 1 = R.$

NYTI:(ii) Find the interval of convergence. IOC $= (0, 2)$

NYTI: Find the Taylor expansion for $f(x) = x^{\frac{1}{2}}$ centered at 4. $= \sum_{0}^{\infty} c_n (x-4)^n$

	n	$f^{(n)}(x)$	$f^{(n)}(4)$	$c_n = \frac{f^{(n)}(4)}{n!}$
+	0	$x^{1/2}$	2	
→ +	1	$\frac{1}{2}x^{-1/2}$	$\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$	
−	2	$\frac{1}{2}(-\frac{1}{2})x^{-3/2}$		
+	3	$\frac{1}{2}(-\frac{1}{2})(-\frac{3}{2})x^{-5/2}$		
−	4	$\frac{1}{2}(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})x^{-7/2}$		
	$\vdots$			
	n	$\frac{(-1)^{n-1}1\cdot3\cdot5\cdots(2n-3)}{2^n}\cdot x^{-\frac{2n-1}{2}}$	$\xrightarrow{x=4} \frac{(-1)^{n-1}1\cdot3\cdot5\cdots(2n-3)}{2^n}\cdot(2^{\cancel{2}})^{-\frac{2n-1}{\cancel{2}}}$ $= \frac{(-1)^{n-1}1\cdot3\cdot5\cdots(2n-3)}{2^{3n-1}}$	

$\boxed{(n \geq 2)}$

$$x^{1/2} = c_0 + c_1(x-4) + \sum_{\boxed{n=2}}^{\infty} c_n (x-4)^n$$

$$= 2 + \frac{1}{4}(x-4) + \sum_{n=2}^{\infty} \frac{(-1)^{n-1}1\cdot3\cdot5\cdots(2n-3)}{2^{3n-1}\cdot n!}(x-4)^n$$

Taylor Polynomials

Let a be a real number and f a function whose Taylor series expansion about a is given by

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots$$

If we consider a partial sum of the power series, we obtain a polynomial. We call

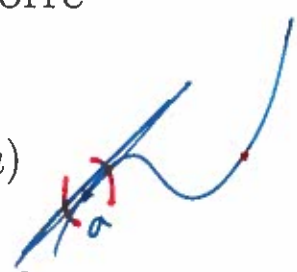
$$T_N(x) = \sum_{n=0}^{N} c_n(x-a)^n = c_0 + c_1(x-a) + \cdots + c_N(x-a)^N$$

the N**th (degree) Taylor Polynomial** of f centered at a.

calc 1

Notice that the first Taylor polynomial corresponds to the linearization of f at a:

$$T_1 = L(x) = c_0 + c_1(x-a) = f(a) + f'(a)(x-a)$$



Taylor polynomial generalizes this by providing polynomial constructed with higher derivatives. We sometimes call the Nth Taylor polynomial the N**th order approximation** of f.

For example, the second order approximation of f at a is given by

$$T_2(x) = f(a) + f'(a)(x-a) + \underbrace{\frac{f''(a)(x-a)^2}{2}}$$

ex. 4. Find the Taylor series of $\boxed{\sin x}$ centered $= \sum_0^\infty c_n (x-\frac{\pi}{2})^n$ at $\frac{\pi}{2}$. Use the Taylor polynomial $\boxed{T_4}$ to estimate the value of $\sin(91°)$.

n	$f^{(n)}(x)$	$f^{(n)}(\frac{\pi}{2})$	$c_n = \frac{f^{(n)}(\frac{\pi}{2})}{n!}$
0	$\sin x$	1	
~~1~~	~~$\cos x$~~	~~0~~	
2	$-\sin x$	-1	
~~3~~	~~$-\cos x$~~	~~0~~	
4	$\sin x$	1	

$$\sin x = 1 - \frac{1}{2!}\left(x-\frac{\pi}{2}\right)^2 + \frac{1}{4!}\left(x-\frac{\pi}{2}\right)^4 - \frac{1}{6!}\left(x-\frac{\pi}{2}\right)^6 + \cdots$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}\left(x-\frac{\pi}{2}\right)^{2n}$$

$$T_4 = 1 - \frac{1}{2!}\left(x-\frac{\pi}{2}\right)^2 + \frac{1}{4!}\left(x-\frac{\pi}{2}\right)^4$$

$$\sin(91°) = \sin\Big(\underbrace{\frac{91\pi}{180}}_{\substack{\shortparallel \\ x}}\Big) \sim 1 - \frac{1}{2!}\left(\frac{\pi}{180}\right)^2 + \frac{1}{4!}\left(\frac{\pi}{180}\right)^4$$

$$\text{error} \le \frac{1}{6!}\left(\frac{\pi}{180}\right)^6$$

ex. Find the $\boxed{\textbf{sum}}$ of the series: $\displaystyle\sum_{0}^{\infty} \frac{(-1)^n \pi^{2n}}{6^{2n}(2n)!}$

$$\cos x = \sum_{0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = \sum_{0}^{\infty} \frac{(-1)^n}{(2n)!} \left(\frac{\pi}{6}\right)^{2n}$$

$$= \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

ex. Compute the $\boxed{10\text{th}}$ derivative of

$$f(x) = \arctan\left(\frac{x^2}{3}\right), \qquad \text{at } x = 0.$$

(hint: Use the Maclaurin series for $f(x)$ to find $f^{10}(0)$.)

$$f(x) = \sum_{0}^{\infty} c_n x^n = \sum \frac{f^{(n)}(0)}{n!} x^n$$

$$c_n = \frac{f^{(n)}(0)}{n!}$$

$$\boxed{f^{(n)}(0) = c_n \cdot n!}$$

$$\tan^{-1}(x) = \sum_{0}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{2n+1}$$

$$\tan^{-1}\left(\frac{x^2}{3}\right) = \sum_{0}^{\infty} (-1)^n \frac{\left(\frac{x^2}{3}\right)^{2n+1}}{2n+1}$$

$$= \sum_{0}^{\infty} \underbrace{\boxed{\frac{(-1)^n}{3^{2n+1} \cdot (2n+1)}}}_{c_n} \cdot x^{\boxed{4n+2}} = 10$$

$$\underline{\underline{n = 2}}$$

$$f^{(10)}(0) = c_{10} \cdot 10!$$

$$= \boxed{\frac{(-1)^2}{3^5 \cdot 5}} \cdot 10!$$

NYTI:

1. Find the Maclaurin Series for $g(x) = \sin(x^2)$.
   Use T_7 to approximate the value of $\int_0^1 \sin(x^2)dx$.

2. Find the Maclaurin series for $f(x) = x\cos x$.
   Write your answer in summation notation.

3. Find the Maclaurin series for $f(x) = \ln(x+5)$ using either the definition of a Maclaurin series or the geometric series.

4. a. Find the **sum** of the series:

$$1 - \ln 2 + \frac{(ln2)^2}{2!} - \frac{(ln2)^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n (\ln 2)^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{[-\ln 2]^n}{n!} = e^{-\ln 2} = \frac{1}{2}$$

b. Find the function with the series $\sum_{n=0}^{\infty} (-1)^n \frac{x^{4n}}{n!}$.

$$= \sum_{n=0}^{\infty} \frac{(-1)^n \cdot (x^4)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-x^4)^n}{n!} = e^{-x^4}$$

5. Power series cover the entire landscape of calculus from limits and derivatives to integrals and approximation.

Limits by Taylor series

ex. Evaluate $\lim_{x \to 0} \frac{x^2 + 2\cos x - 2}{3x^4}$ $\left(\frac{0}{0}\right)$ L'H.

ex. Use the Taylor series for $\sin x$ to verify that $\lim_{x \to 0} \boxed{\frac{\sin x}{x}} = 1$.

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\lim_{x \to 0} \frac{\sin x}{x} = \lim_{x \to 0} \boxed{1} - \underbrace{\frac{x^2}{3!}}_{\to 0} + \underbrace{\frac{x^4}{5!}}_{\to 0} - \underbrace{\frac{x^6}{7!}}_{\to 0} + \cdots = 1$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

$$2\cos x = 2 - x^2 + 2\cdot\frac{x^4}{4!} - 2\cdot\frac{x^6}{6!} + \cdots$$

$$\frac{x^2 + 2\cos x - 2}{3x^4} = \frac{2\cdot\frac{x^4}{4!} - 2\frac{x^6}{6!} + \cdots}{3x^4}$$

$$= \frac{\frac{2}{4!}\cancel{x^4}}{3\cancel{x^4}} - \frac{\frac{2}{6!}x^6}{3x^4} + \cdots$$

$$\lim_{x\to 0} \frac{x^2 + 2\cos x - 2}{3x^4} = \lim_{x\to 0} \frac{1}{36} - \frac{\frac{2}{6!}}{3}x^2 + \cdots$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$0 \quad 0 \quad 0 \quad \cdots$$

$$= \frac{1}{36}.$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+2}}{(2n+1)!}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n (x^2)^{2n+1}}{(2n+1)!}$$

NYTI:

1. Find the Maclaurin Series for $g(x) = \sin(x^2)$.

   Use T_7 to approximate the value of $\int_0^1 \sin(x^2)dx$.

$$\int_0^1 \sin(x^2)dx \sim \int_0^1 \left(x^2 - \frac{x^6}{3!}\right)dx$$

$$= \frac{x^3}{3} - \frac{x^7}{42}\Big|_0^1 = \frac{1}{3} - \frac{1}{42} = \frac{13}{42}$$

2. Find the Maclaurin series for $f(x) = x\cos x$.
   Write your answer in summation notation.

$$x\cos x = x \cdot \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n)!}, \quad R=\infty$$

3. Find the Maclaurin series for $f(x) = \ln(x+5)$ using either the definition of a Maclaurin series or the geometric series.

$$\frac{1}{x+5} = \frac{1}{5} \cdot \frac{1}{1+\frac{x}{5}} = \frac{1}{5} \cdot \frac{1}{1-(-\frac{x}{5})} = \frac{1}{5} \cdot \sum_{n=0}^{\infty} \left(-\frac{x}{5}\right)^n$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{5^{n+1}} \qquad \left|-\frac{x}{5}\right| < 1, \quad |x| < 5, \quad R=5$$

$$\int \frac{1}{x+5}dx = \int \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{5^{n+1}} dx$$

$$\ln(x+5) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{(n+1)\cdot 5^{n+1}} + C$$

let $x=0$: $\ln 5 = 0 + C$, $C = \ln 5$

$$\ln(x+5) = \ln 5 + \sum^{\infty} \frac{(-1)^n x^{n+1}}{(n+1)\cdot 5^{n+1}}, \quad R=5$$

4. a. Find the **sum** of the series:

$$1 - \ln 2 + \frac{(ln2)^2}{2!} - \frac{(ln2)^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n (\ln 2)^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{(-\ln 2)^n}{n!} = e^{-\ln 2} = e^{\ln \frac{1}{2}} = \frac{1}{2}$$

b. Find the function with the series $\sum_{n=0}^{\infty} (-1)^n \frac{x^{4n}}{n!}$.

$$= \sum_{n=0}^{\infty} \frac{(-1)^n (x^4)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-x^4)^n}{n!} = e^{-x^4}$$

5. Power series cover the entire landscape of calculus from limits and derivatives to integrals and approximation.

Limits by Taylor series

<u>ex.</u> Evaluate $\lim_{x \to 0} \frac{x^2 + 2\cos x - 2}{3x^4}$ (see next page.)

<u>ex.</u> Use the Taylor series for $\sin x$ to verify that $\lim_{x \to 0} \frac{\sin x}{x} = 1$.

$$= \lim_{x \to 0} \frac{x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots}{x}$$

$$= \lim_{x \to 0} 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \cdots$$

$$= 1$$

Evaluate $\lim_{x \to 0} \dfrac{x^2 + 2\cos x - 2}{3x^4}$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots$$

$$2\cos x = 2 - x^2 + 2 \cdot \frac{x^4}{4!} - 2 \cdot \frac{x^6}{6!} + 2 \cdot \frac{x^8}{8!} - \cdots$$

$$x^2 + 2\cos x - 2 = \frac{1}{12} x^4 - \frac{2}{6!} x^6 + \frac{2}{8!} x^8 - \cdots$$

$$\frac{x^2 + 2\cos x - 2}{3x^4} = \frac{\frac{1}{12} x^4}{3x^4} - \frac{\frac{2}{6!} x^6}{3x^4} + \frac{\frac{2}{8!} x^8}{3x^4} - \cdots$$

$$= \frac{1}{36} - \frac{2}{3 \cdot 6!} x^2 + \frac{2}{3 \cdot 8!} x^4 - \cdots$$

$$\lim_{x \to 0} \frac{x^2 + 2\cos x - 2}{3x^4} = \lim_{x \to 0} \left(\frac{1}{36} - \underbrace{\frac{2}{3 \cdot 6!} x^2}_{\downarrow\, 0} + \underbrace{\frac{2}{3 \cdot 8!} x^4}_{\downarrow\, 0} - \cdots \right)$$

$\downarrow\, 0 \quad \downarrow\, 0$

$$= \frac{1}{36}$$